Shaw Campbell

campbellshaw8@gmail.com | +27 79 733 6179 | LinkedIn | GitHub | Website

For RiskWorx: I have kept up to date with mathematics, learning mathematics for quant finance that you can see evidence of at Website. I have taken a break from that and am doing rigorous computer science theory now, you can see at GitHub. I tutored my department's only stats course called signals 2. I will pick up advanced stats and finance very fast, Kanishk can vouch for me. My programming skills are far above average, especially in CSharp and Java.

Education

University of Cape Town — Cape Town, South Africa
BSc(Eng) Honours in Electrical & Computer Engineering — *2021 – 2025*

- Graduated with Honours; overall GPA 71%.
- Honours thesis: deep-learning deformable registration of ultrasound to MRI for prostate imaging (PyTorch, VoxelMorph, ANTs); semi-supervised training outperformed unsupervised, with affine pre-alignment and augmentation critical given limited data.

Experience

PSG Financial Services — Cape Town, South Africa
Graduate Software Developer (Intern) — *2026 – Present*

- Build and maintain production web applications (.NET / Angular, TypeScript) on an advice platform used by thousands of financial advisors; write C# end-to-end tests with Playwright.
- Migrated 100+ automated tests from a legacy framework to Playwright using AI coding tools (DeepSeek, OpenCode), cutting regression cycles from months to weeks.

Peralex Electronics — Cape Town, South Africa
Software Engineering Intern — *2025 (5 weeks)*

- Designed and built a Python desktop app for visualising openEMS electromagnetic simulation results across multiple antenna types, within a full Agile cycle.

Electrum Payments — Cape Town, South Africa
Software Development Intern — *2024 (4 weeks)*

- Delivered a feature end-to-end on a live production codebase (Java, Spring Boot, REST APIs) embedded in the engineering team; received a full-time return offer.

University of Cape Town — Cape Town, South Africa
Tutor & Laboratory Assistant, Electrical Engineering & Computer Science — *2023 – 2025*

- Tutored Signals & Systems I/II and introductory Python/Java programming to classes of 300+ students; Control Systems Laboratory Assistant (2023, 2024).

Selected Projects

Multi-Modal Deformable Image Registration — 2026

- Converting messy thesis Jupyter notebooks into a clean, modular, documented Python repository for training image-registration models.

TaskFlow: CPU-Inspired Task Scheduler (C++) — 2025

- Translated a full UML design into a working task scheduler implementing classical CPU scheduling algorithms and OOP design patterns, with CMake.

Extracurricular & Awards

IT & Maintenance Representative, Leo Marquard Hall House Committee — 2023

* Served on a committee overseeing a residence of 400+ students and hundreds of thousands of rands in funds; helped secure winter heater funding for residents.

Deans Merit List Award — 2022 – 2023

UCT Plus Gold Award (for over 60 hours of elected leadership position) — 2023

PSG Financial Services Merit Based Bursary (Full-ride two year bursary) — 2024 – 2025

Skills

Languages: Python, Java, C++, C, C#, TypeScript/JavaScript, HTML/CSS
Tools & Frameworks: TensorFlow, .NET, ASP.NET, Angular, Spring Boot, Playwright, Jupyter, Git, Linux, MS SQL Server, NUnit, Selenium